Matrix gradients and signed channel interactions

Consolidated appendix draft for the coarse information manuscript

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Abstract

We consolidate finite matrix gradient estimates, spectral repair bounds and signed channel examples into one supporting appendix. An operator-valued Cwikel estimate gives an inverse moment bound with explicit counting-measure normalization. Finite-rank repair produces a capped moment criterion that admits zero and negative channels. A separate effective-resistance argument constructs stable signed families with uniform observation rank and penalty, and a small Schur matrix detects coupled instabilities. An explicit torus example shows that individually repairable channels need not remain repairable together by a scalar differentiable observation. The underlying spectral and resistance methods are classical. The statements concern the defined finite gradient form; they establish neither an interacting Yang–Mills form identity nor a continuum mass gap.

Status and provenance. This is a grouped appendix candidate for the existing integrated manuscript, not a separate deposit. Its analytical sources and finite evidence are pinned at repository commit `72baba23`; full identifiers and execution records are in Section 6. The published working paper v6 and its assessment report 154 remain separate artifacts. This new PDF has no inherited assessment score, independent review or Lean certificate. The parent is AIRR working paper ARR-2026-7EPM3V3DP796BBRR version 6; its unchanged PDF SHA256 is

`ccf53e0afc9819c3fc9b2008de9076a60f60d23ef6ac2c4ccd82adca30cbf389`.

1 Actual finite objects and observation scope

Let $D \geq 3$, $n \geq 2$, $h \geq 1$, $V = n^D$, $r = Dh$, and $p = D/2$. On $X = (\mathbb{Z}/n\mathbb{Z})^D$ use the counting inner product. For $u : X \to \mathbb{R}^h$, define

$$du(x) = (u(x + e_i) - u(x))_{i=1}^{D}, \qquad Q_A(u) = \sum_x du(x)^T A(x) du(x), \quad L_A = d^* A d,$$

where $A(x)$ is a real symmetric $r \times r$ matrix. A channel is one local coefficient eigenvector, counted with multiplicity. Distinct matrices need not commute. All derivatives have unit spacing. A physical derivative nd requires an explicit rescaling of every energy and threshold.

The number $N_{<c}(L_A)$ counts all eigenvalues strictly below $c > 0$, including negative and zero ones, with ordinary real multiplicity. Spectral coordinates on this low space give a real observation C and penalty

$$\Phi(y) = \sum_{\lambda<c} (c - \lambda) y_\lambda^2, \qquad Q_A(u) + \Phi(Cu) \geq c\|u\|^2. \tag{1}$$

This is an existence result for an actual, generally nonlocal observation. It does not construct the unresolved local rectangular observation. For signed forms, $c\|y\|^2$ alone need not be sufficient.

2 The positive matrix estimate

Put

$$\alpha_D = \frac{D}{2}\left(\frac{D}{D-2}\right)^{D-1}\left(\frac{3}{4}\right)^{D}.$$

Theorem 1. *If every $A(x)$ is positive definite, then*

$$N_{<c}(L_A) \le h + \alpha_D c^p \sum_x \operatorname{tr} A(x)^{-p}. \tag{2}$$

Proof. On the mean-zero space H_0, write $L_0 = d^*d$ and $J = dL_0^{-1}$. For any $b \in H_0$, the flow $j_* = AdL_A^{-1}b$ has $d^*j_* = b$. Every other feasible flow $j = j_* + k$ has $d^*k = 0$, and expansion gives

$$\langle j, A^{-1}j\rangle = \langle b, L_A^{-1}b\rangle + \langle k, A^{-1}k\rangle.$$

Taking the actual feasible flow Jb proves $L_A^{-1} \le J^*A^{-1}J$. No coefficients commute in this argument.

The unitary counting Fourier transform on r components has $L_1(\ell_2^r) \to L_\infty(\ell_2^r)$ norm $V^{-1/2}$. The symbol of J is $z(k)/\lambda(k)$ with $z_i(k) = e^{2\pi i k_i/n} - 1$ and $\lambda(k) = 4\sum_i \sin^2(\pi k_i/n)$; set $J(0) = 0$. It is rectangular $r \times h$. Embed $\mathbb{R}^h$ isometrically in $\mathbb{R}^r$ by I and use the square symbol $G(k) = J(k)I^*$. Thus $\|G(k)\| = \lambda(k)^{-1/2}$ for $k \neq 0$. Writing $\mathcal{F}_r$ for the counting transform, the actual factorization is $J = \mathcal{F}_r^* M_G \mathcal{F}_r I$. Centered frequency representatives b satisfy $\lambda(k) \ge 16|b|^2/n^2$. Counting the cube containing $0 < |b| < n/(4t)$ gives

$$\sup_{t>0} t^D \#\{k : \|G(k)\| > t\} \le (3/4)^D V.$$

When the radius is at most one the set is empty; otherwise $(2\lfloor R\rfloor + 1)^D \le (3R)^D$ suffices.

Apply Frank's operator-valued Cwikel theorem [1, Theorem 3.1] with $q = D$, $f = A^{-1/2}$, this square multiplier and counting measures. Its constant is $\frac{D}{2}(\frac{D}{D-2})^{D-1}V^{-1}$, so the displayed frequency V cancels exactly once. It bounds the number of singular values of $A^{-1/2}J$ strictly above s by $\alpha_D s^{-D}\sum_x \operatorname{tr} A(x)^{-p}$. The right factor $\mathcal{F}_r I$ is an isometry and cannot increase singular-value counts. Apply min-max to the inverse comparison at threshold $1/c$ on H_0, then add the h constant modes. Complexification of a real symmetric matrix preserves multiplicities. □

Input and interpretation. The operator-valued estimate is classical; this proof fixes its finite normalization, padding and actual form dictionary. It is not a new general Cwikel theorem. A matrix arithmetic mean alone does not imply its inverse moment hypothesis. For the formal flat cochain application one may set $h = D(N_c^2 - 1)$, but an interacting Hessian or weighted curl/divergence form must first be proved equal or comparable to this Q_A.

3 Spectral repair and the capped budget

Theorem 2. *Let $A(x)$ now be arbitrary real symmetric matrices. In local orthonormal eigenbases choose a set S of channels containing all nonpositive ones. Let P_S be their orthogonal projection and $R(S) = \operatorname{rank}(P_S d)$. Then*

$$N_{<c}(L_A) \le \min\left\{hV,\ h + R(S) + \alpha_D c^p \sum_{j\notin S} \lambda_j^{-p}\right\}. \tag{3}$$

Writing ν_0 for the total multiplicity of nonpositive local eigenvalues, one consequently has

$$N_{<c}(L_A) \le \min\{hV,\ h + F_c(A)\}, \quad F_c(A) = \nu_0 + \sum_{\lambda_j>0} \min\{1, \alpha_D c^p \lambda_j^{-p}\}. \tag{4}$$

Proof. Replace selected eigenvalues by T larger than zero and all of them. The resulting field A_T is positive definite and

$$L_{A_T} - L_A = d^* P_S (T\mathrm{I} - A) P_S d \geq 0, \quad \operatorname{rank}(L_{A_T} - L_A) = R(S).$$

The rank equality follows because $T\mathrm{I} - A$ is strictly positive on $\operatorname{im} P_S$. Finite-rank min-max gives $N_{<c}(L_A) \leq N_{<c}(L_{A_T}) + R(S)$: intersect the low space of L_A with the kernel of the update and the orthogonal complement of the low space of L_{A_T}. A dimension larger than $R(S)$ would give a nonzero vector whose identical quadratic forms are both strictly below and at least $c\|u\|^2$. Theorem 1 and $T \to \infty$ prove (3). For (4), use $R(S) \leq |S|$ and select all positive channels with $\alpha_D c^p \lambda_j^{-p} \geq 1$ as well. □

A strict extension of the full moment criterion. Let $m \geq 1$ and $T_m = \max\{1, V/(1+m)^p\}$. If $F_c(A) \leq KT_m$, then $N_{<c}(L_A) \leq (h+K)T_m$. Every former SPD budget

$$\frac{1}{rV} \sum_x \operatorname{tr}((mA(x)^{-1})^p) \leq M$$

implies this capped budget, with $K = 2^p \alpha_D c^p r M$. Conversely choose $m = n^2$, one actual diagonal edge channel $\epsilon = n^{-4}$, and all others m. Now $T_m = 1$ and $F_c(A) \leq 1 + \alpha_D c^p r$, whereas the normalized full inverse moment is at least n^{2D}/r and diverges. These bounds are derived for this explicit family; no new moment bound is assumed.

Sparse contamination and its limitation. Selecting every channel with $\lambda < m$ gives

$$N_{<c}(L_A) \leq h + R_m + \alpha_D c^p r V m^{-p}, \qquad R_m = \operatorname{rank}(\mathbf{1}_{(-\infty,m)}(A)d).$$

An actual selected-channel count $k_m \leq KT_m$ suffices since $R_m \leq k_m$. If $A(x) \geq m\mathrm{I}$ outside a specified set B, then $k_m \leq r|B|$. Neither condition follows from an arithmetic mean. This particular prescribed-threshold count criterion does not subsume every full moment family: $A = (m/2)\mathrm{I}$ has normalized inverse moment 2^p but every channel lies below m. The two particular sufficient criteria are incomparable; the general capped criterion (4) does include the old one.

Actual rank and unavoidable costs. For diagonal direction/component channels the selected rows are graph incidences. If κ_a counts all components, including isolated vertices, of the selected-edge graph for component a, then $R(S) = \sum_a (V - \kappa_a)$: its kernel consists precisely of fields constant on those components. Parallel rows at $n = 2$ do not change that kernel. This formula is not applied to rotated local eigenchannels. Isolating q nonadjacent vertices by zeroing their incident edges gives at least $q+1$ kernel modes. Replacing these zeros by $0 < \epsilon < c/(2D)$ yields a q-dimensional strict low subspace of delta functions. The defect-count cost cannot simply be deleted.

In (1), a generic bound is $K \leq c + 4Db$ with $b = \max_x \|A_-(x)\|$, since $\|du\|^2 \leq 4D\|u\|^2$. One negative channel $-B$ with all other channels m, at $n \geq 3, h = 1$, has the trial vector $v = \delta_y - \delta_x$ on that edge and

$$\frac{Q_A(v)}{\|v\|^2} = -2B + (2D-1)m.$$

Any normalized linear observation with penalty bounded by $K\|y\|^2$ therefore needs $K \geq c + 2B - (2D-1)m$. Sparsity alone does not bound penalty.

4 Stable signed families from actual resistance

This section is independent of the Cwikel input. Its mechanism is classical effective resistance and finite-rank congruence [2, 3]. Let e be one actual direction/component row and $v = d^*e$. With all coefficients m except this one set to $-B$, $B \geq 0$,

$$H = mL_0 - (m + B)vv^*, \qquad L_0 = d^*d.$$

On H_0 let $G = L_0^{-1}$. Translations and coordinate permutations make every scalar row leverage equal. Their sum is $\operatorname{tr}(dGd^*) = V - 1$, so

$$\rho = v^*Gv = \frac{V - 1}{DV}. \tag{5}$$

Components decouple; the same value holds for every h. The DV rows are counted individually even at $n = 2$.

Proposition 3. *The exact threshold is $B_* = m(DV/(V - 1) - 1)$. On H_0, H is positive definite for $B < B_*$, has one zero direction at equality, and has exactly one negative direction above it. If $m = n^2$, $B = \theta m$ and $0 \leq \theta < D - 1$, its mean-zero gap is at least*

$$g = 16(D - 1 - \theta)/D > 0.$$

For every $0 < c \leq g$, the normalized mean observation $C(u) = V^{-1/2} \sum_x u(x) \in \mathbb{R}^h$ and $\Phi(y) = c\|y\|^2$ give coercivity with optimal rank h, uniformly in n.

Proof. Congruence by $G^{1/2}$ gives $m\mathrm{I} - (m + B)ww^*$ with $\|w\|^2 = \rho$, proving the phase claim. The equality witness is Gv. Cauchy–Schwarz in the L_0 energy gives $|v^*u|^2 \leq \rho\langle u, L_0u\rangle$, hence $H \geq [m - (m + B)\rho]L_0$ on H_0; equality holds on Gv. For the stated scaling the coefficient is at least $m(D - 1 - \theta)/D$. The actual first nonconstant Fourier eigenvalue of L_0 is $4\sin^2(\pi/n) \geq 16/n^2$. This proves the gap. Decompose into constants and H_0 to prove the displayed observation inequality. The constant kernel forces rank at least h. □

For $\theta > 0$, negative coefficient depth grows as θn^2, yet the penalty is uniformly c. A bounded-depth hypothesis is sufficient for the generic estimate, not necessary for every stable signed family. The actual form is positive despite an indefinite coefficient field; this does not contradict the fixed-m, $B \to \infty$ penalty obstruction above.

5 Coupled channels and a nonlinear observation obstruction

Choose q actual rows of d as Z, and $W = \operatorname{diag}(w_j) > 0$. The modified form is exactly $H = mL_0 - Z^*WZ$. Define

$$\Gamma = ZGZ^*, \qquad \mathcal{C} = mW^{-1} - \Gamma, \quad N_0 = h(V - 1).$$

Schur elimination of either positive diagonal block in $\left[\begin{smallmatrix} mL_0 & Z^* \\ Z & W^{-1} \end{smallmatrix}\right]$ on $H_0 \oplus \mathbb{R}^q$ proves

$$n_-(H) = n_-(\mathcal{C}), \quad n_0(H) = h + n_0(\mathcal{C}), \quad n_+(H) = N_0 - q + n_+(\mathcal{C}). \tag{6}$$

Indeed the two congruences give the identity

$$\operatorname{inertia}(H|_{H_0}) + (q, 0, 0) = (N_0, 0, 0) + \operatorname{inertia}(\mathcal{C}/m).$$

No disjoint-cycle or independent-row assumption occurs. In particular, H is positive semidefinite with exactly h constant zeros if and only if $\mathcal{C}$ is positive definite. General signed matrix criteria preexist in [4]; our Γ uses the positive reference inverse.

Example 4. *Take $D = 3, n = 2, h = 1, m = 4$, and change the first two direction rows at one vertex separately or together to coefficient $-84/11$. Then*

$$\Gamma = \begin{pmatrix} 7/24 & 5/48 \\ 5/48 & 7/24 \end{pmatrix}, \quad w = 128/11, \quad w/m = 32/11.$$

The diagonal follows from (5)*; the off-diagonal Walsh sum is $(1/8)(1/2 + 1/3) = 5/48$, with both forward orientations counted. Each single channel is below its critical magnitude $68/7$ and has full inertia $(7, 0, 1)$. Together the reduced matrix has diagonal $5/96$ and off-diagonal $-5/48$, so its eigenvalues are $-5/96, 5/32$. Equation* (6) *gives joint inertia $(6, 1, 1)$.*

Each single form has mean-zero gap at least $16(1 - (32/11)(7/24)) = 80/33 > 1$. The same scalar mean observation with $\Phi(y) = |y|^2$ repairs either single form at $c = 1$. The joint form cannot be repaired by any scalar observation $F : \mathbb{R}^8 \to \mathbb{R}$ differentiable at zero with $F(0) = 0$ and $0 \le \Phi(z) \le K|z|^2$ near zero. Its negative eigenvector and constant vector span a two-dimensional nonpositive space. The derivative of F has a nonzero kernel vector v there. Then $F(tv) = o(t)$ and $\Phi(F(tv)) = o(t^2)$, whereas $Q_A(tv) \le 0$, contradicting coercivity at $c = 1$ for small nonzero t.

This refutes the shortcut from individual to joint control, not the full matrix resistance criterion or any theorem retaining a disjoint-cycle hypothesis. The q-row perturbation also gives $N_{<c}(H) \le h + q$ for $0 < c \le 16m/n^2$. Extra zero or negative modes provide the necessary rank lower bound in (6); they do not alone bound the penalty coefficient.

6 Reproduction and limitations

The source repository is THE ERIKSSON PROGRAMME PR186, stacked on PR185. All paths below are under `reproducibility/matrix-moment-sweep-20261106/`. Full source for this consolidation is pinned at

`72baba2356c209a75c96cd8e96b758620dbf972d`.

Units 162 and 163 each used two distinct own CPU allocations, four actual fresh sparse Git clones at one pinned source, and eight normal/optimized Python processes. Outputs matched within each unit. Both runtimes were deleted and disconnected after evidence export. Full stdout, stderr, exit sentinels, Git logs, exact source bytes and manifests are in the corresponding `evidence162-*.zip` and `evidence163-*.zip` archives.

Unit	Executed source prefix	Fixture cases	Mutations per process
162	`f644c093`	four	seven types
163	`378e879b`	four	seven types across four cases

The 163 fixtures are $(D, n, h) = (3, 2, 1), (3, 3, 1), (3, 3, 2), (4, 2, 2)$. They check actual Green and leverage identities, below/equal/above phase inertia, equality witnesses, gap and mean-observation coercivity, and coupled pair inertia. Unit 162 covers zero and near-zero stars, one large negative channel and rationally rotated local eigendirections. Every acceptance check raises explicitly on failure in both ordinary and `-O` execution. There are 28 fixture-mutation pairs per 163 process, not 28 distinct types. The two163 recipe times were 53.83 and 29.98 seconds; requested-to-release times were 179.71 and 102.74 seconds. Billed compute units were not exposed.

Finite replay checks these finite matrices, not the universal analytical statements. Arbitrary q, unequal weights, the all-size capped-family example and the nonlinear derivative consequence are analytical arguments, not extra executed fixtures. Earlier 159 diagnostics used source

downloads rather than fresh Git clones; their original limitation remains in their own records. No Lean compilation certifies this appendix. A full independently prepared audit of the imported Fourier adapter and the application form dictionary remains necessary before broader claims.

7 Model assessments and review context

This appendix was prepared and checked analytically in the continuing Codex session that prepared its sources and diagnostics. The application does not expose the exact backend model identity for that preparation. There is no claim of independence, blinding, human peer review or editorial acceptance. This exact new PDF has no external model assessment and no numerical score. An assessment of an earlier PDF does not transfer to this artifact.

The retained objections are substantive: the Cwikel mechanism and Schur principles are classical, so independent originality is unestablished; the observation from spectral coordinates is generally nonlocal; sparse-count and full-moment criteria must not be conflated; a diagonal graph-rank formula does not cover rotated eigenchannels; negative depth is neither universally dispensable nor universally necessary; the full off-diagonal interaction must be retained; finite diagnostics do not prove all-size statements; and no interacting physical form equality has been supplied. These limits are part of the result, not editorial acceptance conditions represented as satisfied.

The original positive critical rectangular construction remains open. The common-refinement route remains stopped after three unsuccessful units. The recorded PR29 baseline stays 20/41 with TermSource 0. This appendix adds supporting finite mathematical results and useful counterexamples without changing those records. It is a draft for consolidation, not a deposit, acceptance or new verified public release.

References

[1] R. L. Frank, *Cwikel's theorem and the CLR inequality*, arXiv:1206.3325v1 (2012), Theorem 3.1. `https://arxiv.org/abs/1206.3325v1`.

[2] D. Zelazo and M. Bürger, *On the definiteness of the weighted Laplacian and its connection to effective resistance*, IEEE CDC (2014), Theorem III.3. `https://arxiv.org/abs/1408.2187`.

[3] D. A. Spielman, *Spectral Graph Theory Lecture 14*, 15 October 2018, Claim 14.6.2. `https://www.cs.yale.edu/homes/spielman/561/lect14-18.pdf`.

[4] W. Chen, J. Liu, Y. Chen, S. Z. Khong, D. Wang, T. Başar, L. Qiu and K. H. Johansson, *Characterizing the positive semidefiniteness of signed Laplacians via effective resistances*, IEEE CDC (2016), pp. 985–990. `https://people.kth.se/~kallej/papers/network_cdc16chen.pdf`.